CE3

Project Introduction

CE3.1

Project Name	ENSE3020 Group Project: Bicycle Odometer
Project Duration	09/2020 to 10/2020
Project Organization	The University of Western Australia
My Title	Electronics Engineer

CE3.2

A device installed on a bicycle to measure and record the distance traveled by the user. Besides distance, it provides the rider with information on cycling efficiency, route planning, and maintenance of the bike. With advancements in technology, bicycle odometers are evolving to become more intelligent and multifunction. Future development for bicycle odometers involves integrating more sensors, enhancing data analysis capabilities, providing deeper integration with mobile devices, and providing personalized cycling recommendations.

1. Product front view



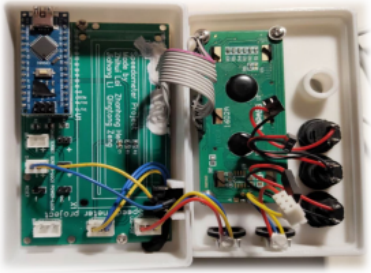
2. What's in the Box



3. Installation Bottom View



4. Inner Schema (without Battery)



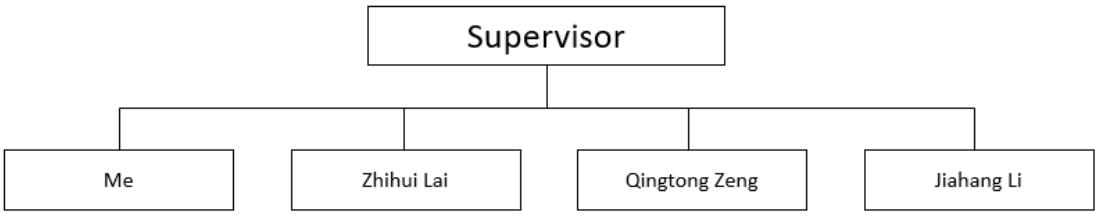


5. Installation Front View

Project Background

CE3.3

Under my supervisor's guidance, I clearly outlined the project milestones in the opening report and maintained close communication with my mentor and teammates through chat software. Additionally, I developed a comprehensive project implementation plan, clarified the team's roles, established effective communication mechanisms and risk prevention systems, and regularly backed up all project documents to a dedicated hard drive to ensure data security.



CE3.4

The main objective of this project is to design a bicycle odometer.

CE3.5

The main tasks of this project included software design, hardware design, algorithm research, and project budget analysis. I worked with my teammates, and we divided the tasks according to our respective strengths. As a result, my primary responsibilities were focused on software and hardware design. I designed the software and hardware components, ensuring they worked seamlessly together to achieve the project's objectives.

Project Activities

CE3.6

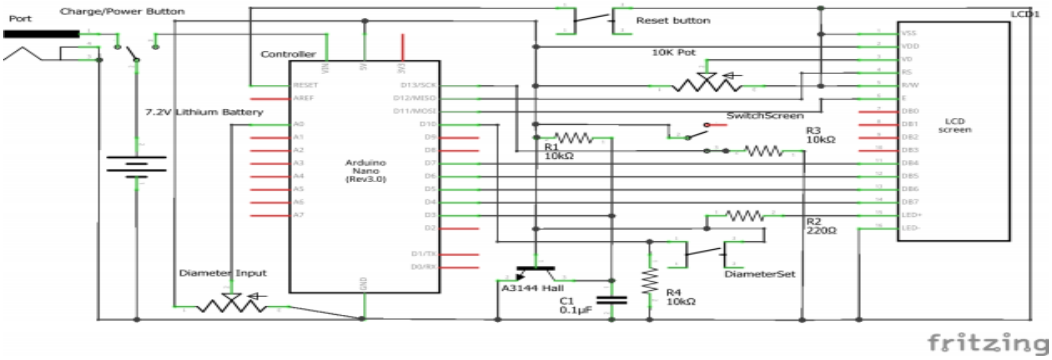
- In order to display the current speed and odometer readings, I first calculated the circumference of the tire. Then, I limited the range of the potentiometer signal from (0-1023) to (400-800), ensuring that the tire diameter was set only within the range of standard road bicycle tires (400 mm to 800 mm). Next, to calculate speed and distance, I determined the time required for the tire to complete one full rotation by calculating the time interval between the signals from two Hall sensors. After obtaining these parameters, I used the relevant formulas to calculate the speed, applying the conversion factor of 3.6 km/h = 1 m/s, and ultimately derived the speed in kilometers per hour (km/h). At the same time, I calculated the total distance traveled using the distance formula.
- After calculating these basic parameters, I began to restrict the range of each parameter. To ensure that the current speed would refresh in real-time when the cycling stopped, I set a condition: if no signal input was detected for more than 2 seconds, the speed value would be set to zero. Additionally, to eliminate signal jitter, I required the Arduino controller to only

read the sensor signal if the time interval between signals exceeded 60 milliseconds. This approach allowed me to ensure the accuracy of the speed measurement. I also defined the speed range as between 3 km/h and 120 km/h. Given that typical bicycle speeds are much lower than 120 km/h, this setup fully met the requirements and effectively measured speed and distance. Furthermore, I referenced relevant literature and standards to determine that the battery voltage should be within the range of 7V to 12V to ensure stable system operation. Considering environmental factors, I also decided to power the device using the Arduino's Vin pin or by connecting a 5V power supply through a USB-mini port.

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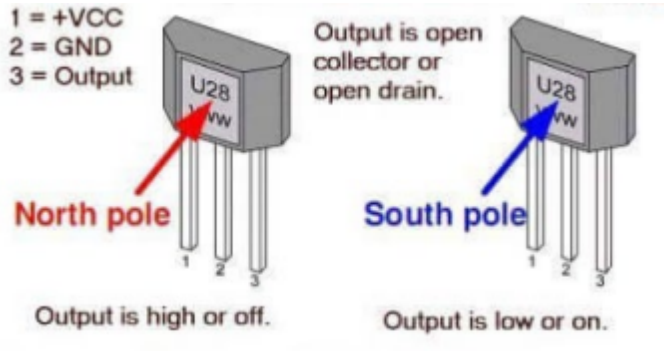
When designing the circuit diagram, I used Fritzing software to create the schematic. The main electronic components in the design included the Arduino, an LCD display, resistors, capacitors, and Hall sensors.

- I configured five input interfaces for the Arduino controller. First, I used a potentiometer as an analog input to adjust the bicycle tire diameter through the A0 pin. Then, I set up three digital input interfaces, connecting buttons to the D10, D13, and RESET pins. These buttons included the diameter setting button, screen switching button, and reset button. I added pull-down resistors (R3 and R4) for each button, while the reset button was connected through the Arduino's internal resistor. The pull-down resistors provided an initial logic low state for the buttons, preventing excessive current when pressing the button connected to 5V, thus protecting the Arduino controller from damage. Then, I connected the Hall sensor's signal to the D3 pin to provide a digital signal for calculating speed and distance.



- In choosing the Hall effect sensor, I reviewed relevant literature and noted the fact that the application involved was not considering linearly accurate properties. We agreed, after discussions with my teammates, to use the A3144 type Hall sensor with a digital output. I used the resistor labeled R1 as a pull-up and the capacitor, in this case, C1, for filtering any noise that might disturb the signal. The pull-up resistor caused the Hall sensor to be in a high logic state on the input, thus preventing extreme amounts of current from passing through and damaging the Arduino controller if found applied over the Hall sensor. More so, special regard was given to the fact that for the Hall sensor to output a VCC signal, the south pole of

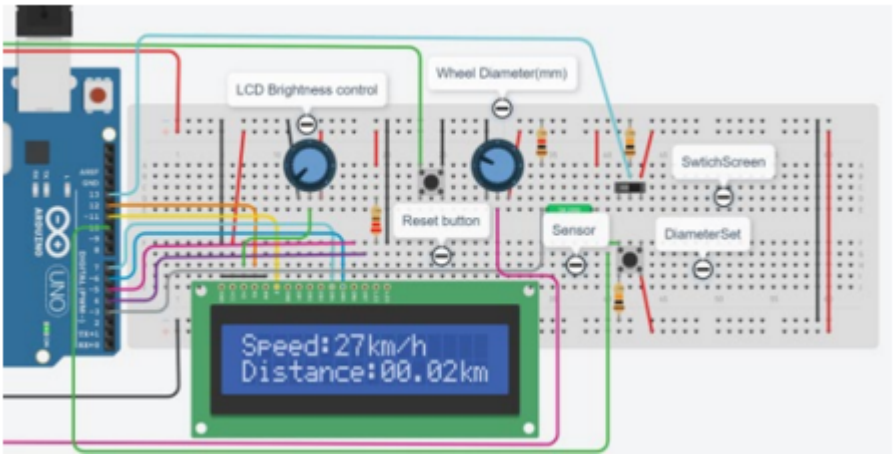
the magnet must face the front of the Hall sensor.

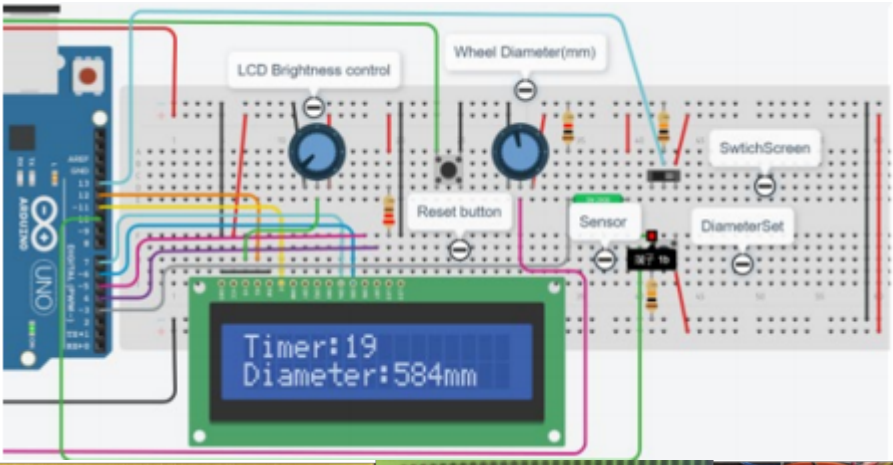


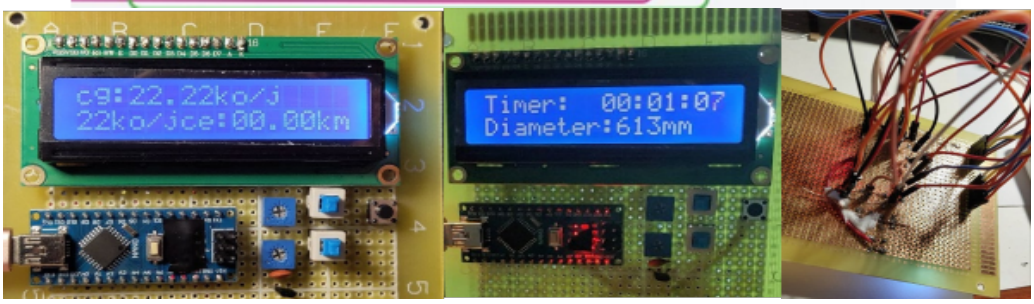
- For the LCD display, I chose the LCD 1602 model and followed the standard wiring procedures. The register select pin (RS) was connected to D12, and the read/write pin (R/W) was connected to ground to ensure it was always in write mode. The enable pin (E) was connected to D11, and the four data pins (DB4-DB7) were connected to D7-D4. Additionally, I controlled the display contrast pin (Vo) using a 10k potentiometer and connected the positive terminal of the LED backlight (LED+) to 5V through a 220Ω resistor (R2), while the negative terminal (LED-) was connected to ground.
- The battery voltage range was from 7V to 12V, and I connected the battery to the Arduino's Vin pin to power the device. Additionally, I set up a power button to control the device's on/off function, but this button did not control the 5V power supplied through the USB-mini port.

CE3.8

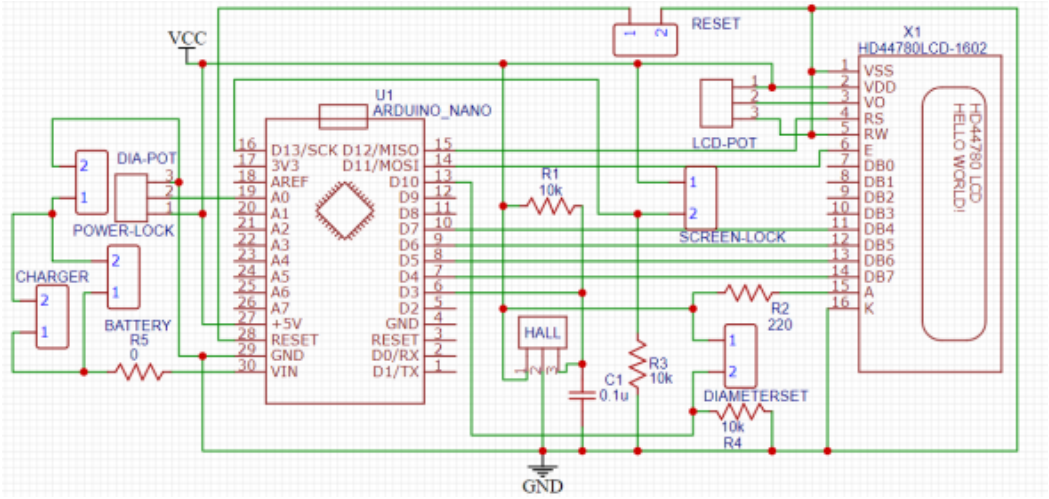
- While assisting my teammates with circuit simulation using TinkerCAD, I was unable to directly simulate the Hall sensor. As a result, I used an inclinometer switch as a substitute for the Hall sensor. I purchased a breadboard and other necessary components, and then soldered them according to the design specifications.

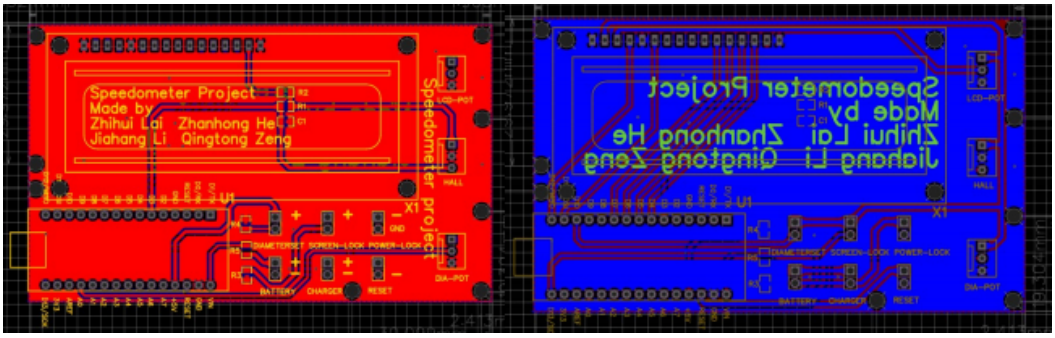




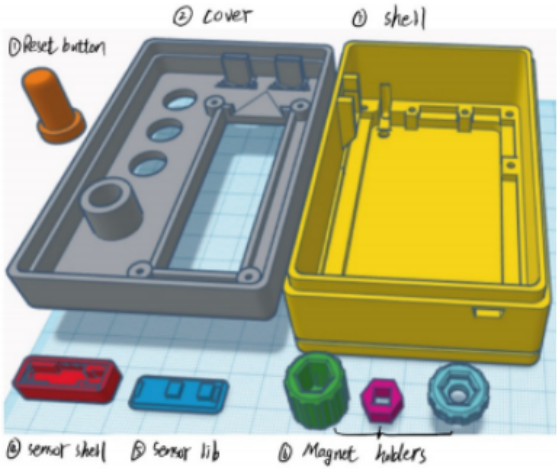


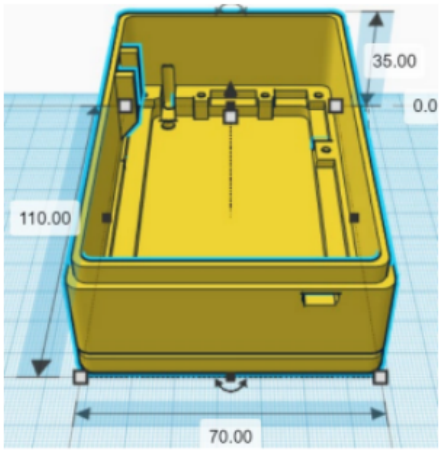
- Based on the successful testing of the circuit, I further optimized the design. To simplify the soldering and connections, I innovatively replaced the buttons and battery with JST-2 pin connectors and the potentiometer with a JST-3 pin connector in the EasyEDA schematic. This not only simplified the PCB soldering process but also made it easier to expand the connections of key components, facilitating their installation into the enclosure. I then placed the components in appropriate positions based on the principle of ease of soldering and wiring, ensuring enough spacing to generate PCB paths. Next, I used the auto-routing feature to have the PCB layout automatically fill in the circuit paths. To enhance the current-carrying capacity of the ground plane, I covered both the top and bottom layers with copper and connected them to the ground. Then, I added screw holes at the edges of the board to allow it to be securely fixed within the enclosure.

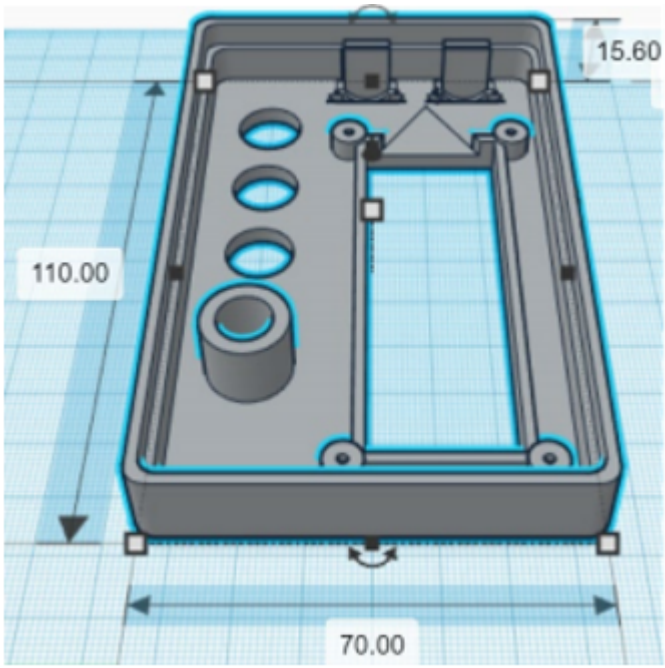




- After completing the PCB design, I proceeded with the product's exterior design. I divided the appearance into three sections: the first section was the main box, which was designed to house the PCB board, external components, and the battery; the second section was the sensor box, intended to house the Hall sensor and make it easy to install on the bicycle frame; the third section was the magnet holder, which was designed to securely attach a 5mm cubic magnet to the wheel using screws.







CE3.9

During the hardware design process, I first selected the components to ensure that all design requirements were met. For the battery, I considered two installation options: the first option used four 3V button cells (CR2032), providing a total voltage of 12V, while the second option used two 3.7V lithium batteries (18650), providing a total voltage of 7.4V, with the added benefit of being rechargeable. Based on the components we purchased, I decided to design two versions of the battery configuration and confirmed that both options would effectively support the normal operation of the product. Next, I chose the 1602 LCD display to ensure it would meet the product's display needs. I also selected 5mm cubic magnets and the A3144 digital Hall sensor to enable the measurement of speed and distance. To ensure the circuit's stability, I used 10kΩ and 220Ω resistors in 0805 package size and a 0.1μF capacitor in 0805 package size to filter out noise. For the buttons, I designed two latching buttons: one for power control and the other for switching the screen display. Additionally, I designed two non-latching buttons for resetting and setting the tire diameter. I chose the Arduino Nano as the main control chip due to its compact size, which was ideal for the project's requirements. For the potentiometer, I selected two 10kΩ dial potentiometers to input the tire diameter and adjust the LCD contrast. To ensure the stability of the entire circuit, I designed a double-sided PCB measuring 100mm x 60mm with a thickness of 1.6mm, using green surface treatment, with green paint covering the hole locations, and both sides were coated with solder for easy soldering. After completing the circuit board design, I continued with the product's exterior design and component connections. Considering the different battery options, I designed a corresponding battery case for each version, one using four 3V button cells and the other using two 3.7V lithium batteries.

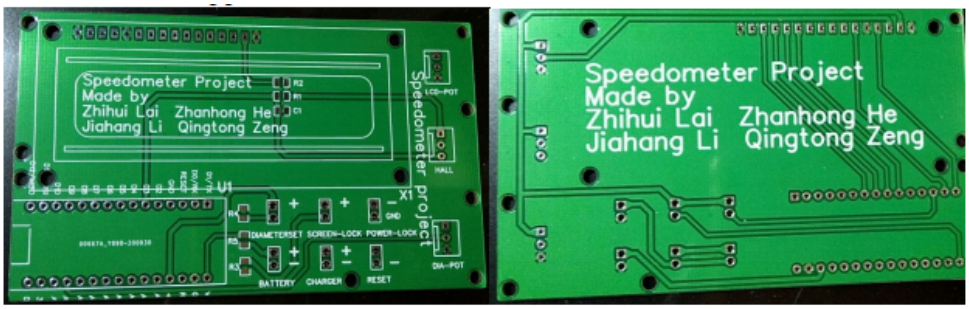


FIGURE 8: FINISHED PCB BOARD WITH MEMBER NAME ON THE FRONT (L) AND BACK SIDE (R)



FIGURE 9: TESTING DESIGNED PRODUCT WITHOUT SHELL

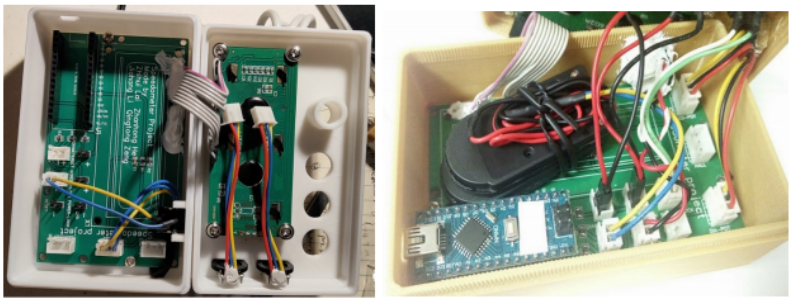


FIGURE 10: USING SCREW TO INSTALL PCB AND LCD ON SHELL/LIP, WELDING WIRE UP COMPONENTS WITH JST CONNECTOR

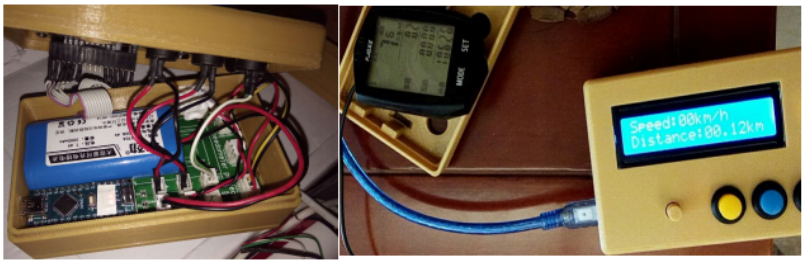


FIGURE 11: WHEN EVERYTHING IS READY, FINALLY INSTALL THE BATTERY -- TRIAL DONE!

CE3.10

The design of this project did not take into account the variation in natural light during different times of the day. However, during the subsequent optimization process, I planned to integrate a light-dependent resistor (LDR) to continuously monitor the ambient light intensity. Based on the sensor data, the system would then automatically adjust the display brightness for better visibility and energy efficiency.

Project Summary

CE3.11

I paid strict attention to the project timeline, ensuring the processes of both design and debugging were completed on time. While testing the circuit, I made sure that I carefully checked connections between the various components and the operation of each component itself. I carried out thorough literature reviews on the relevant technical literature to ensure that my design was in line with the cutting-edge technological trends and industry standards. Besides, special attention was paid to the recyclability of the battery and components. The 18650 lithium-ion battery chosen, was a rechargeable type, which not only did away with the waste of disposable batteries but allowed proper recycling after the end of its lifespan.